HOCHSCHULE
RAVENSBURG-WEINGARTEN
UNIVERSITY
OF APPLIED SCIENCES

Electronic Circuit Design

Lecture notes

Chapter 7: Active Filters

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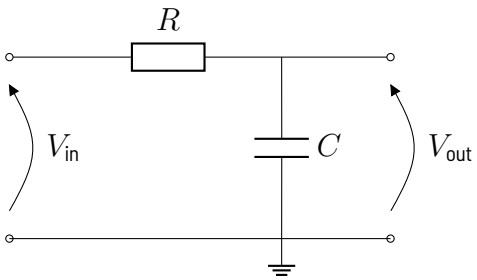
7 Active Filters

7.1 Passive RC Filters

A filter lets some frequencies pass and attenuates others. In a measurement system the wanted signal is usually slow, while interference (mains hum at 50 Hz, switching noise, radio frequencies) is faster. A low-pass filter in front of the analog-to-digital converter removes it. It has a second, more fundamental task: a sampling system with sample rate f_s cannot distinguish a signal at frequency f from one at $f \pm k f_s$. Everything above $f_s/2$ must be removed *before* sampling; this is the anti-aliasing filter (chapter 9 explains aliasing in detail).

We start with the passive RC filter and the tools to describe it: complex transfer function, magnitude and phase, decibels and the Bode plot. Then we add op-amps (chapter 6) to get first-order and second-order active filters, and finally design the 2.5 Hz anti-aliasing low-pass of the capstone task. Further reading: [2] (chapters on filters), [1] (chapter 6), [5] (active filter design) and [3].

RC low-pass



- low frequency: the capacitor is open, $V_{\text{out}} = V_{\text{in}}$
- high frequency: the capacitor is a short, $V_{\text{out}} \to 0$
- a frequency-dependent voltage divider with the impedance $Z_C = \dfrac{1}{j\omega C}$

RC low-pass

$$\underline{H}(j\omega) = \frac{\underline{V}_{\text{out}}}{\underline{V}_{\text{in}}} = \frac{1/(j\omega C)}{R + 1/(j\omega C)} = \frac{1}{1 + j\omega RC}$$

Magnitude, phase and cutoff frequency

- with $f_c = \dfrac{1}{2\pi RC}$: $\quad \underline{H} = \dfrac{1}{1 + jf/f_c}$
- magnitude (amplitude ratio) and phase:

First-order low-pass

$$|\underline{H}| = \frac{1}{\sqrt{1 + (f/f_c)^2}} \qquad \varphi = -\arctan(f/f_c) \qquad f_c = \frac{1}{2\pi RC}$$

- at $f = f_c$: $|\underline{H}| = 1/\sqrt{2} = 0.707$, $\varphi = -45°$
- far above f_c: $|\underline{H}| \approx f_c/f$ – ten times the frequency, one tenth of the amplitude

For a sine input the output of a linear circuit is a sine of the same frequency. Its amplitude is multiplied by $|\underline{H}|$ and its phase is shifted by φ. The cutoff frequency f_c (also corner frequency or -3 dB frequency) is the frequency at which the real part and the imaginary part of the denominator are equal; there the output power is half the power at low frequencies. The time constant $\tau = RC$ and the cutoff frequency are two views of the same thing: $f_c = 1/(2\pi\tau)$.

Decibels

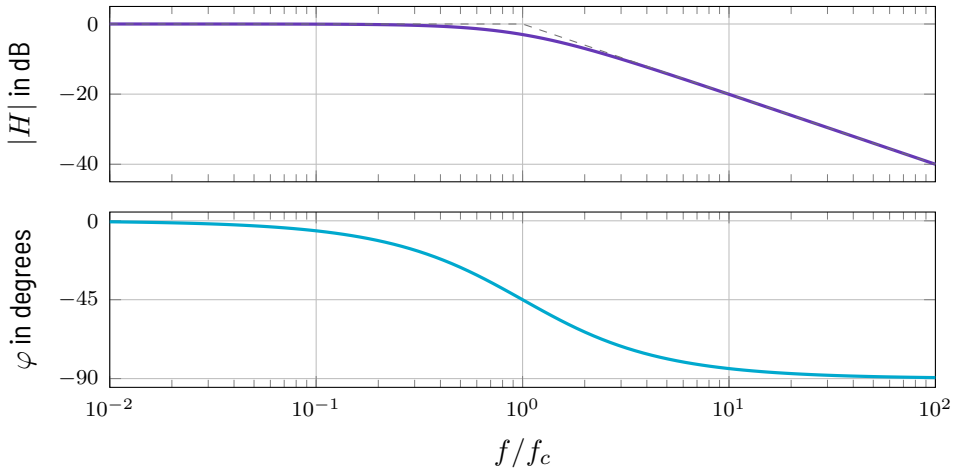
- gains and attenuations span many decades: a logarithmic measure is convenient

Definition 7.1 The gain in decibels is $|H|_{\text{dB}} = 20 \log_{10} |H|$ (voltage or current ratios).

$\lvert H\rvert$	dB	$\lvert H\rvert$	dB
1	0 dB	2	6 dB
$1/\sqrt{2} = 0.707$	−3 dB	10	20 dB
0.5	−6 dB	100	40 dB
0.1	−20 dB	1000	60 dB
0.01	−40 dB	10^5	100 dB

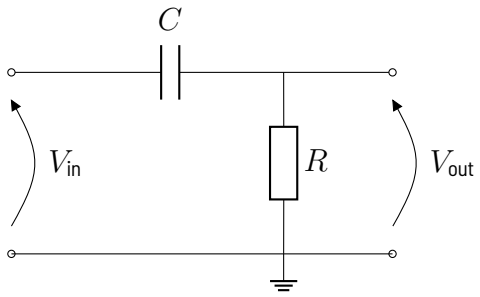
- gains of cascaded stages **multiply**; in dB they **add**
- a decade is a factor 10 in frequency, an octave a factor 2

Bode plot of the RC low-pass



- logarithmic frequency axis, magnitude in dB: the curve becomes two straight asymptotes
- 0 dB below f_c, -20 dB per decade above; the corner is at f_c (-3 dB)

RC high-pass



First-order high-pass

$$\underline{H} = \frac{jf/f_c}{1 + jf/f_c} \qquad |\underline{H}| = \frac{f/f_c}{\sqrt{1 + (f/f_c)^2}} \qquad \varphi = 90° - \arctan(f/f_c)$$

- blocks DC (coupling capacitor), 20 dB per decade below f_c
- same cutoff frequency $f_c = 1/(2\pi RC)$, phase $45°$ at f_c

RC filters in Falstad

- $R = 1.6\,\text{k}\Omega$, $C = 100\,\text{nF}$: $f_c = 995\,\text{Hz}$; three copies driven with $1\,\text{V}$ at $100\,\text{Hz}$, $1\,\text{kHz}$, $10\,\text{kHz}$

	$100\,\text{Hz}$	$1\,\text{kHz}$	$10\,\text{kHz}$
low-pass amplitude	$0.995\,\text{V}$	$0.705\,\text{V}\ (-45°)$	$0.099\,\text{V}$
high-pass amplitude	$0.100\,\text{V}$	$0.709\,\text{V}\ (45°)$	$0.995\,\text{V}$

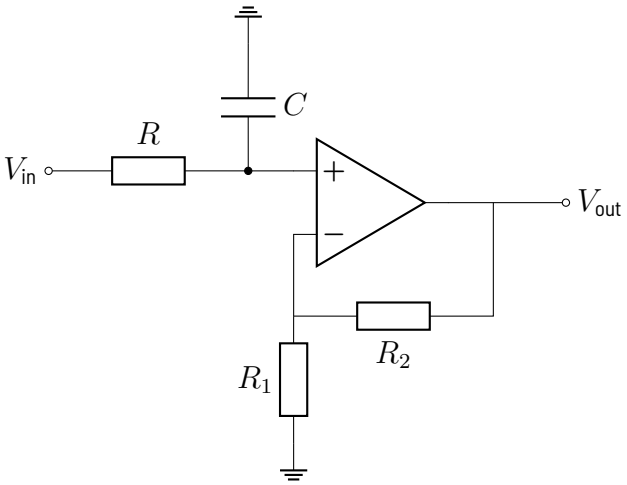
▶ Falstad: `examples/filters/filter_rc_lowpass.txt` ▶ Falstad: `examples/filters/filter_rc_highpass.txt`

- limits of passive filters: the load changes f_c (it is in parallel to C or R), no gain, steep filters need inductors
- remedy: an op-amp buffers the filter and adds gain $\rightarrow$ **active filter**

7.2 First-Order Active Filters

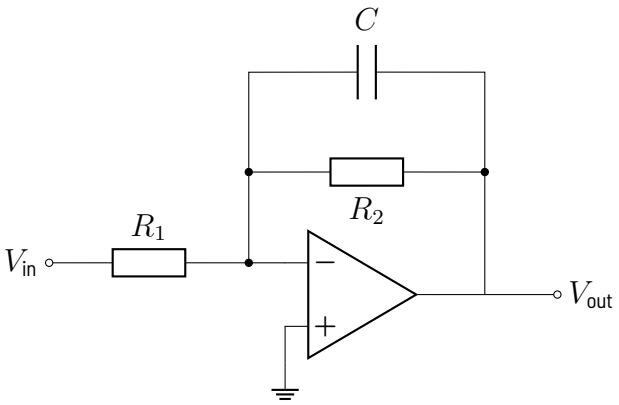
An op-amp turns the RC filter into an active filter: the filter is no longer loaded by the following stage, and it can have a gain. In the simplest case the RC network sits in front of a non-inverting amplifier; alternatively the capacitor is placed into the feedback path of an inverting amplifier. The cutoff frequency is still $1/(2\pi RC)$ for the resistor and capacitor that form the filter.

Non-inverting first-order low-pass



- RC low-pass, then a non-inverting amplifier (or a follower for $G = 1$)
- $\underline{H} = \left(1 + \frac{R_2}{R_1}\right) \frac{1}{1 + j\omega RC}$, $f_c = \frac{1}{2\pi RC}$
- the op-amp input does not load C; the output can drive the next stage

Inverting first-order low-pass

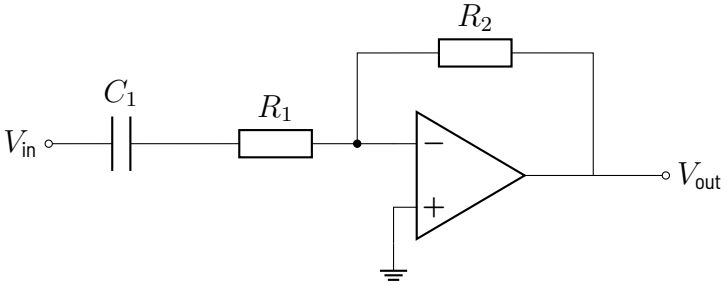


- feedback impedance $Z_2 = R_2 \parallel \frac{1}{j\omega C}$, gain $-Z_2/R_1$

Inverting first-order low-pass

$\underline{H} = -\frac{R_2}{R_1} \frac{1}{1 + j\omega R_2 C}$ DC gain $-\frac{R_2}{R_1}$, $f_c = \frac{1}{2\pi R_2 C}$

First-order high-pass (inverting)



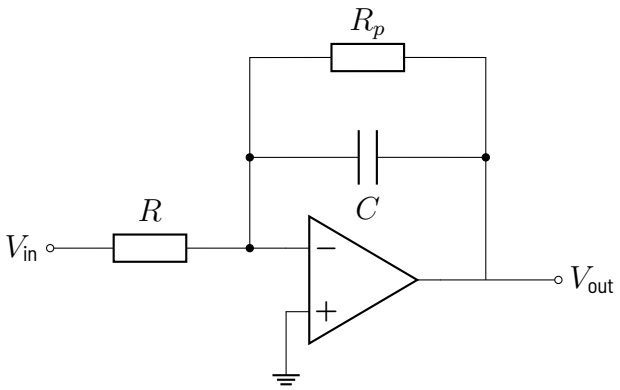
Inverting first-order high-pass

$$\underline{H} = -\frac{R_2}{R_1}\,\frac{j\omega R_1 C_1}{1 + j\omega R_1 C_1} \qquad \text{HF gain } -\frac{R_2}{R_1}, \quad f_c = \frac{1}{2\pi R_1 C_1}$$

- Falstad: all three circuits with $f_c = 995$ Hz, input 0.1 V at 1 kHz
  - non-inverting LP ($G = 2$): 0.141 V; inverting LP (-10): 0.705 V; inverting HP (-10): 0.709 V

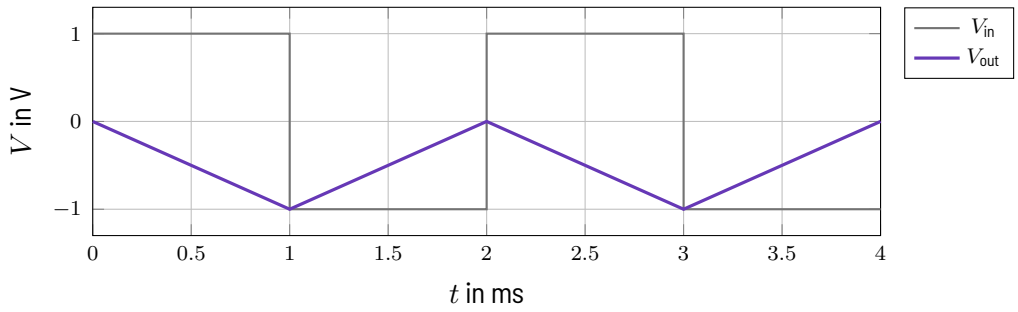
▶ Falstad: `examples/filters/filter_active_first_order.txt`

Integrator



- the current V_{in}/R charges C: $\quad V_{\text{out}}(t) = -\dfrac{1}{RC}\int V_{\text{in}}\,dt, \quad \underline{H} = -\dfrac{1}{j\omega RC}$
- -20 dB per decade at all frequencies; DC gain ∞: any offset drives it into saturation
- practical integrator: R_p limits the DC gain to $-R_p/R$ (it is then a low-pass with $f_c = 1/(2\pi R_p C)$)

Integrator: square wave in, triangle out



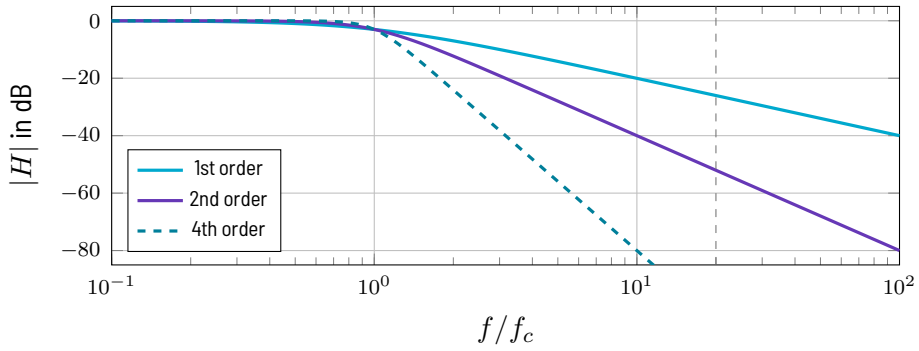
- $R = 10$ kΩ, $C = 100$ nF, $R_p = 1$ MΩ; $V_{\text{in}} = \pm 1$ V, 500 Hz
- slope $V_{\text{in}}/(RC) = 1000$ V/s: 1 V in 1 ms, triangle 1.0 V peak-to-peak
- the triangle starts between 0 V and -1 V; R_p centres it slowly ($\tau = R_p C = 0.1$ s)

▶ Falstad: `examples/filters/filter_integrator.txt`

7.3 Second-Order Filters

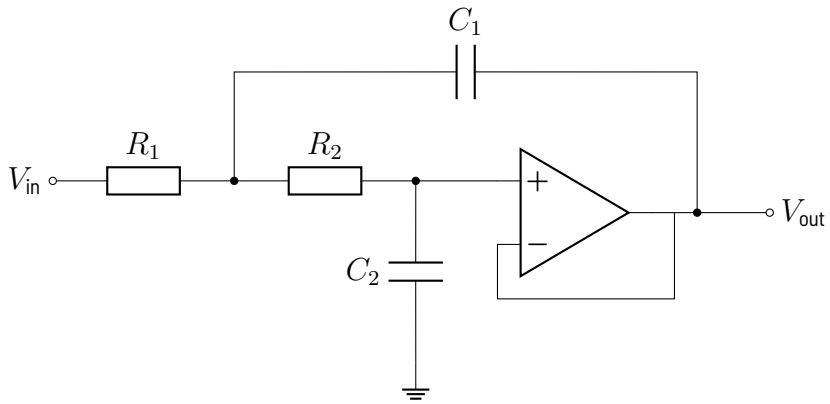
A first-order filter falls by only 20 dB per decade. If the interference is only one decade above the signal, a first-order filter attenuates it by a factor of ten – often not enough. Cascading RC stages helps little: passive RC stages load each other and the corner becomes soft. Active second-order filters give 40 dB per decade with one op-amp, and a sharp corner whose shape is set by the quality factor Q. The most popular circuit is the Sallen-Key filter [5, 2].

Filter order and attenuation



- Butterworth filter of order n: $|\underline{H}| = 1/\sqrt{1+(f/f_c)^{2n}}$; slope $-20n$ dB per decade
- dashed line: $f = 20\ f_c$ (50 Hz for $f_c = 2.5$ Hz): -26 dB, -52 dB, -104 dB

Sallen-Key low-pass (unity gain)



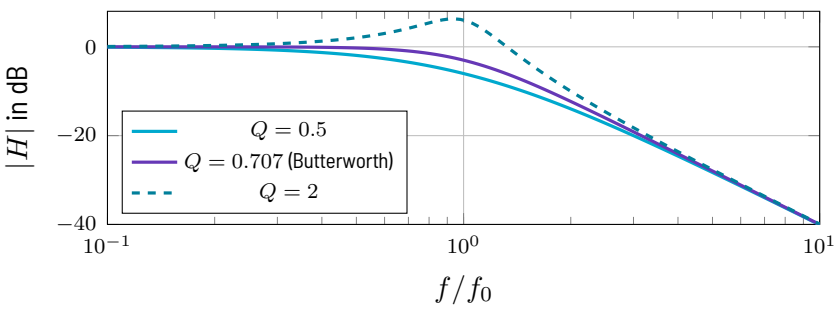
- the follower makes $V_{\text{out}} = V_+$; C_1 feeds the output back to the middle node (positive feedback near f_0)
- $\underline{H} = \dfrac{1}{1 + j\omega C_2(R_1+R_2) - \omega^2 R_1 R_2 C_1 C_2}$

Natural frequency and quality factor

- comparing with $1/(1 + j\frac{f}{Qf_0} - (f/f_0)^2)$ gives f_0 and Q:

Unity-gain Sallen-Key low-pass

$$f_0 = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}} \qquad Q = \frac{\sqrt{R_1 R_2 C_1 C_2}}{C_2\,(R_1+R_2)} \overset{R_1=R_2}{=} \frac{1}{2}\sqrt{\frac{C_1}{C_2}}$$



- $|\underline{H}(f_0)| = Q$; Butterworth $Q = 1/\sqrt{2}$: maximally flat, -3 dB at f_0; higher Q: peaking and ringing

Design of a Butterworth Sallen-Key low-pass

- capacitors come in coarse steps: choose them first, then calculate the resistors

Design card: Unity-gain Sallen-Key low-pass, $Q = 0.707$

1. Choose $R_1 = R_2 = R$. Then $Q = 0.707$ requires $C_1 = 2\,C_2$ (C_1 goes to the output).
2. Choose C_2 from the E6 series, $C_1 \approx 2C_2$ (e.g. 10 nF and 22 nF: $Q = 0.74$).
3. Calculate $R = \dfrac{1}{2\pi f_0 \sqrt{C_1 C_2}}$ and round to the E24 or E96 series.
4. Check f_0 and Q with the rounded values. Use film or C0G capacitors (stable, low leakage).

- example $f_0 = 1$ kHz: $C_2 = 10$ nF, $C_1 = 22$ nF, $R = 10.7$ kΩ (E96): $f_0 = 1003$ Hz, $Q = 0.742$

Capacitors are available in coarser steps than resistors, so they are chosen first and the resistors are calculated. Exact Butterworth behaviour is rarely needed; Q between 0.65 and 0.75 changes the response near f_0 by less than 1 dB. Higher-order filters are built as a cascade of second-order stages (and one first-order stage for odd orders) with tabulated values of f_0 and Q for each stage [2].

First order vs. second order in Falstad

	1 kHz	10 kHz
first order RC ($f_c = 995$ Hz)	0.705 V	0.099 V (-20 dB)
Sallen-Key ($f_0 = 1003$ Hz, $Q = 0.742$)	0.744 V	0.0101 V (-40 dB)

- input 1 V amplitude; one decade above the corner the second-order filter attenuates ten times more
- near f_0 both behave similarly

▶ Falstad: `examples/filters/filter_sallen_key.txt`

7.4 The Anti-Aliasing Filter of the Capstone

In the capstone the amplified bridge signal (0 to 4.00 V, chapter 6) is sampled by a sample-and-hold circuit with $f_s = 5$ Hz and converted by a 3-bit flash ADC. The sensor signal itself changes slowly. The problem is the 50 Hz interference of 5 mV in series with the Pt100: it is a differential signal (4.9 mV at the bridge tap), so the instrumentation amplifier amplifies it by the full gain of 107.6 to an amplitude of 0.52 V. This section designs the low-pass filter that removes it, with the values used in the capstone chapter: $R = 6.8$ kΩ, $C = 10$ µF.

Why filter before sampling? (preview of chapter 9)

- sampling theorem (Nyquist, Shannon [10]): only frequencies below $f_s/2$ are reproduced correctly
- with $f_s = 5$ Hz: $f_s/2 = 2.5$ Hz $\Rightarrow$ cutoff of the low-pass $f_c \approx 2.5$ Hz
- a component above $f_s/2$ is not removed by sampling – it appears at a wrong, low frequency (**aliasing**)
  - $50\,\text{Hz} = 10 \cdot f_s$: the samples see the interference always at the same phase – a constant, random-looking error
  - if the mains frequency drifts to 50.1 Hz, the error becomes a slow 0.1 Hz wave in the data
- after sampling nothing can separate the alias from the signal: the filter must act **before** the S&H

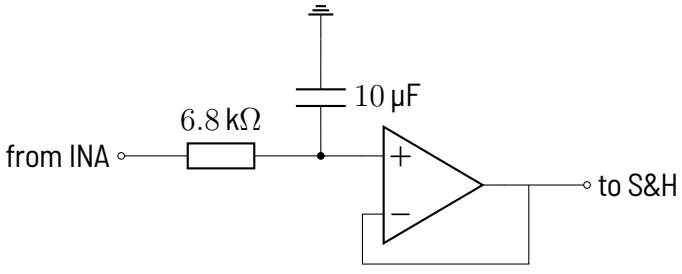
How much attenuation do we get at 50 Hz?

- first order, $f_c = 2.5$ Hz: $|\underline{H}(50\,\text{Hz})| = \dfrac{1}{\sqrt{1+(50/2.5)^2}} = \dfrac{1}{\sqrt{401}} = 0.050 \;\hat{=}\; -26$ dB
- second order (Butterworth): $\dfrac{1}{\sqrt{1+(50/2.5)^4}} = 0.0025 \;\hat{=}\; -52$ dB
- capstone filter $6.8\,\text{k}\Omega$, $10\,\mu$F: $f_c = 2.34$ Hz, $|\underline{H}| = 1/\sqrt{1+(50/2.34)^2} = 0.047 \;\hat{=}\; -26.6$ dB
- interference before the filter: $4.875\,\text{mV} \cdot 107.6 = 0.52$ V

	amplitude	in LSB (0.5 V)	
no filter	0.52 V	1 LSB	useless
first order, 2.34 Hz	25 mV	0.05 LSB	sufficient
Sallen-Key, 2.55 Hz	1.4 mV	0.003 LSB	large margin

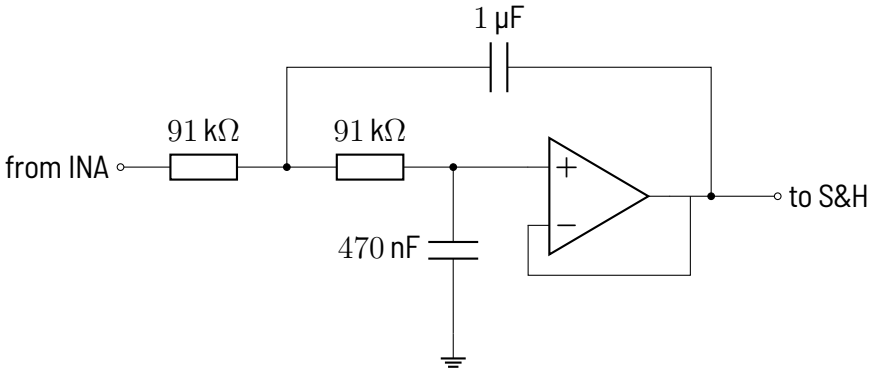
For a 3-bit converter with 4 V full scale one LSB is 0.5 V, and the quantization error is ± 0.25 V. A residual of 25 mV is far below LSB$/2$, so the first-order filter of the capstone is sufficient. A second-order filter costs only one more capacitor and gives a large margin, for example for a higher-resolution ADC: with 10 bits (LSB = 3.9 mV) the first-order filter leaves about 6 LSB of interference, the second-order filter less than half an LSB.

First-order anti-aliasing filter with E-series parts



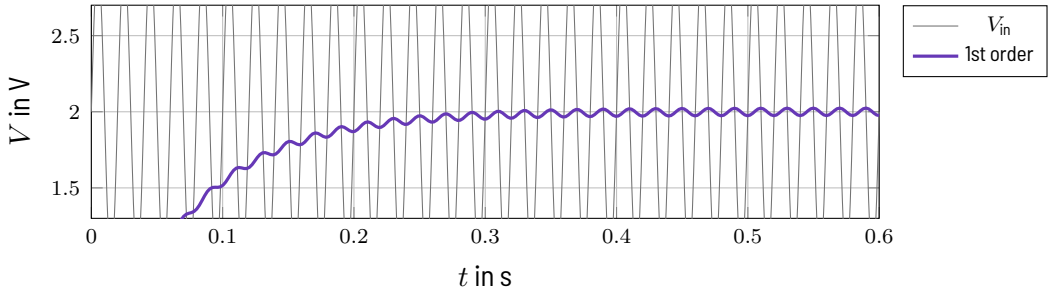
- $RC = 1/(2\pi \cdot 2.5\,\text{Hz}) = 63.7$ ms; choose $C = 10\,\mu$F $\Rightarrow R = 6.37\,\text{k}\Omega$
- round so that $f_c \le f_s/2$: $6.8\,\text{k}\Omega$ (E12) $\Rightarrow f_c = 2.34$ Hz, -26.6 dB at 50 Hz
- also usable: $62\,\text{k}\Omega$ with $1\,\mu$F ($f_c = 2.57$ Hz, slightly above $f_s/2$, still -26 dB at 50 Hz)
- the follower decouples the filter from the S&H input
- step response: $\tau = 68$ ms; settling to $1\,\%$ takes $4.6\,\tau = 0.31$ s

Second-order version (Sallen-Key)



- $C_1 = 1\,\mu\text{F}, C_2 = 470\,\text{nF}$: $Q = \frac{1}{2}\sqrt{1/0.47} = 0.729$
- $R = 1/(2\pi \cdot 2.5\,\text{Hz} \cdot \sqrt{1\,\mu\text{F} \cdot 470\,\text{nF}}) = 92.8\,\text{k}\Omega \rightarrow$ E24 $91\,\text{k}\Omega$: $f_0 = 2.55\,\text{Hz}$

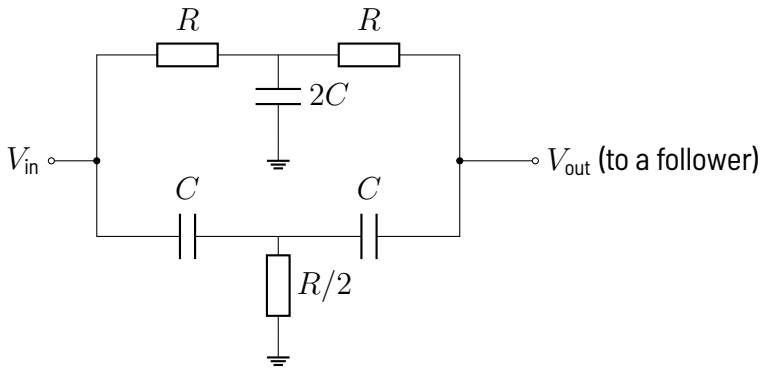
Both filters in Falstad



- input: 2 V signal + 0.52 V, 50 Hz interference (= 4.875 mV $\times$ 107.6)
- after settling: both outputs 2.00 V DC; remaining 50 Hz: 25 mV (1st order), 1.4 mV (Sallen-Key)

▶ Falstad: `examples/filters/filter_antialias.txt`

Alternative: twin-T notch at 50 Hz



- zero transmission at $f_0 = 1/(2\pi RC)$: $R = 31.8\,\text{k}\Omega, C = 100\,\text{nF} \Rightarrow 50\,\text{Hz}$
- Falstad: 50 Hz reduced from 1 V to 0.45 mV; but 5 Hz is also attenuated (0.927)
- needs precise parts, removes only one frequency – it does **not** replace the anti-aliasing low-pass

▶ Falstad: `examples/filters/filter_twin_t.txt`

Design card

Design card: Anti-aliasing low-pass

1. Cutoff frequency from the sample rate: $f_c \leq f_s/2$ (capstone: $f_s = 5$ Hz, $f_c = 2.34$ Hz).
2. Required attenuation: interference amplitude at the ADC input compared with LSB/2; with $f/f_c = 20$ a first-order filter gives -26 dB, a second-order filter -52 dB (capstone: $0.52\,\text{V} \cdot 0.047 = 25\,\text{mV} < 250\,\text{mV}$).
3. First order: $RC = 1/(2\pi f_c)$; choose C, calculate R, round to the E-series so that $f_c \leq f_s/2$ (capstone: 6.8 k, $10\,\mu$F). Buffer the output.
4. Second order: unity-gain Sallen-Key with $C_1 \approx 2C_2$, $R = 1/(2\pi f_0\sqrt{C_1 C_2})$.
5. Check: settling time of a step ($\approx 5\tau$) against the sample period, and headroom of the amplifier before the filter (signal + interference must not clip).

7.5 Exercises

Exercise 7.1: RC low-pass

An RC low-pass has $R = 1.6\,\text{k}\Omega$ and $C = 100$ nF.

a) Calculate the cutoff frequency.

b) Calculate magnitude (also in dB) and phase at 100 Hz, 1 kHz and 10 kHz.

c) What changes if a load of $1.6\,\text{k}\Omega$ is connected to the output?

d) Check b) in Falstad.

Solution 7.1

a) $f_c = 1/(2\pi \cdot 1.6\,\text{k}\Omega \cdot 100\,\text{nF}) = 995$ Hz.

b) 100 Hz: $f/f_c = 0.1005$, $|H| = 1/\sqrt{1.0101} = 0.995$ (-0.04 dB), $\varphi = -5.7°$.
1 kHz: $f/f_c = 1.005$, $|H| = 0.705$ (-3.0 dB), $\varphi = -45.2°$.
10 kHz: $f/f_c = 10.05$, $|H| = 0.099$ (-20.1 dB), $\varphi = -84.3°$.

c) The load is in parallel to C. Thevenin: $V_{\text{th}} = V_{\text{in}}/2$, $R_{\text{th}} = 0.8\,\text{k}\Omega$: the DC gain drops to 0.5 and f_c doubles to 1.99 kHz. A follower avoids this.

d) ▶ Falstad: `examples/filters/filter_rc_lowpass.txt`
amplitudes 0.995 V, 0.705 V (lagging by $45°$, i.e. 125 µs) and 0.099 V.

Exercise 7.2: First-order active filters

a) Design an inverting low-pass with DC gain -10, input resistance $10\,\text{k}\Omega$ and $f_c = 1$ kHz.

b) Calculate the output amplitude for 0.1 V at 1 kHz with C rounded to the E-series.

c) Design a non-inverting low-pass with $G = 2$ and the same f_c, using $C = 10$ nF.

d) Design an inverting high-pass with HF gain -10, $R_1 = 10\,\text{k}\Omega$, $f_c \approx 1$ kHz.

e) Check b) to d) in Falstad (input 0.1 V, 1 kHz).

Solution 7.2

a) $R_1 = 10\,\text{k}\Omega$, $R_2 = 100\,\text{k}\Omega$, $C = 1/(2\pi \cdot 100\,\text{k}\Omega \cdot 1\,\text{kHz}) = 1.59\,\text{nF}$.

b) $C = 1.6\,\text{nF}$ (E24): $f_c = 995\,\text{Hz}$; $0.1\,\text{V} \cdot 10 \cdot 0.705 = 0.705\,\text{V}$.

c) $R = 1/(2\pi \cdot 1\,\text{kHz} \cdot 10\,\text{nF}) = 15.9\,\text{k}\Omega \to 16\,\text{k}\Omega$; gain stage $R_1 = R_2 = 10\,\text{k}\Omega$. Output: $0.1\,\text{V} \cdot 2 \cdot 0.705 = 0.141\,\text{V}$.

d) $R_2 = 100\,\text{k}\Omega$, $C_1 = 1/(2\pi \cdot 10\,\text{k}\Omega \cdot 1\,\text{kHz}) = 15.9\,\text{nF} \to 16\,\text{nF}$: $f_c = 995\,\text{Hz}$, output $0.1\,\text{V} \cdot 10 \cdot 0.709 = 0.709\,\text{V}$.

e) ▶ Falstad: `examples/filters/filter_active_first_order.txt`
`Vout_a` $= 0.141$ V, `Vout_b` $= 0.705$ V, `Vout_c` $= 0.709$ V.

Exercise 7.3: Integrator

An integrator has $R = 10\,\text{k}\Omega$, $C = 100\,\text{nF}$ and $R_p = 1\,\text{M}\Omega$ parallel to C. The input is a square wave of ± 1 V and 500 Hz, starting with $+1$ V at $t = 0$, $V_{\text{out}}(0) = 0$.

a) Calculate the slope of the output voltage and sketch $V_{\text{out}}(t)$ for the first two periods.

b) Calculate the peak-to-peak value of the triangle.

c) What is the purpose of R_p? What happens without it if the op-amp has an offset of 1 mV?

d) Above which frequency does the circuit behave as an integrator?

e) Check a) and b) in Falstad.

Solution 7.3

a) $dV_{\text{out}}/dt = -V_{\text{in}}/(RC) = -1\,\text{V}/1\,\text{ms} = -1000\,\text{V/s}$ while $V_{\text{in}} = +1$ V. Triangle: $0 \to -1$ V in the first 1 ms, back to 0 V at 2 ms, and so on.

b) 1 V per half period: 1.0 V peak-to-peak.

c) R_p limits the DC gain to $-R_p/R = -100$. Without it, 1 mV offset causes a current of 0.1 µA that charges C at 1 V/s: the output saturates after some seconds. With R_p: DC error 0.1 V.

d) Above $f_c = 1/(2\pi R_p C) = 1.6$ Hz; at 500 Hz the error is negligible.

e) ▶ Falstad: `examples/filters/filter_integrator.txt`
the output reaches about -1 V after 1 ms (-0.98 V in the simulation, R_p and the time step cost a little), the triangle has 1.0 V peak-to-peak and drifts slowly towards a symmetric position.

Exercise 7.4: Sallen-Key low-pass

Design a unity-gain Sallen-Key low-pass with $f_0 = 1$ kHz and Butterworth characteristic.

a) Start with $C_2 = 10\,\text{nF}$ and $R_1 = R_2$. Choose C_1 from the E6 series and calculate R (E96).

b) Calculate Q and f_0 with the chosen values.

c) Calculate the amplitude for 1 V input at 1 kHz and 10 kHz. Compare with a first-order filter.

d) Check c) in Falstad.

Solution 7.4

a) $Q = 0.707$ needs $C_1 = 2C_2 = 20$ nF; E6: 22 nF.
$R = 1/(2\pi \cdot 1\,\text{kHz} \cdot \sqrt{22\,\text{nF} \cdot 10\,\text{nF}}) = 10.73\,\text{k}\Omega \to 10.7\,\text{k}\Omega$.

b) $Q = \frac{1}{2}\sqrt{22/10} = 0.742$; $f_0 = 1/(2\pi \cdot 10.7\,\text{k}\Omega \cdot 14.83\,\text{nF}) = 1003$ Hz.

c) $|H| = 1/\sqrt{(1-x^2)^2 + (x/Q)^2}$ with $x = f/f_0$. 1 kHz: $x = 0.997$, $|H| = 0.744$ (0.744 V; first order: 0.705 V). 10 kHz: $x = 9.97$, $|H| = 1/\sqrt{98.4^2 + 13.4^2} = 0.0101$ (−40 dB; first order: 0.099, −20 dB).

d) ▶ Falstad: `examples/filters/filter_sallen_key.txt`
`VSK_1kHz` = 0.744 V, `VSK_10kHz` = 10.1 mV,
`V1st_1kHz` = 0.705 V, `V1st_10kHz` = 0.099 V.

Exercise 7.5: Capstone: anti-aliasing low-pass

The capstone samples with $f_s = 5$ Hz. The amplifier has a gain of 107.6 and its output carries a 50 Hz interference (4.875 mV at the bridge tap). The ADC has 3 bits and 4 V full scale.

a) Choose the cutoff frequency of the anti-aliasing filter and justify it.

b) Design a first-order filter (follower behind an RC) with $C = 10$ µF and an E12 resistor.

c) Calculate the attenuation at 50 Hz in dB for an ideal first-order and second-order filter with $f_c = 2.5$ Hz, and for your filter.

d) Calculate the remaining interference at the ADC input with your filter and compare it with LSB/2.

e) How long does the filter output need to settle within 1 % after a step of the sensor signal?

f) Check the result in Falstad (signal 2 V DC with 0.52 V at 50 Hz).

Solution 7.5

a) Nyquist: only frequencies below $f_s/2 = 2.5$ Hz can be sampled without aliasing, so $f_c \le 2.5$ Hz. The temperature changes much more slowly than that.

b) $R = 1/(2\pi \cdot 2.5\,\text{Hz} \cdot 10\,\mu\text{F}) = 6.37\,\text{k}\Omega \to 6.8\,\text{k}\Omega$ (rounded up, so that f_c stays below $f_s/2$): $f_c = 1/(2\pi \cdot 6.8\,\text{k}\Omega \cdot 10\,\mu\text{F}) = 2.34$ Hz. (62 kΩ with 1 µF, $f_c = 2.57$ Hz, slightly above $f_s/2$, would also be acceptable.)

c) Ideal first order: $1/\sqrt{1+20^2} = 0.0499$, −26 dB; second order: $1/\sqrt{1+20^4} = 0.0025$, −52 dB. With $f_c = 2.34$ Hz: $1/\sqrt{1+(50/2.34)^2} = 0.047$, −26.6 dB.

d) Before the filter: $4.875\,\text{mV} \cdot 107.6 = 0.52$ V; after it: $0.52\,\text{V} \cdot 0.047 = 25\,\text{mV} < \text{LSB}/2 = 250$ mV (0.05 LSB). Sufficient.

e) $\tau = RC = 68$ ms; 1 %: $t = \tau \ln 100 = 4.6\,\tau = 0.31$ s, i.e. about 1.5 sample periods.

f) ▶ Falstad: `examples/filters/filter_antialias.txt`
after about 0.6 s `Vout_1st` shows 2.00 V with 25 mV of 50 Hz ripple; the Sallen-Key output (91 kΩ, 1 µF, 470 nF) leaves only 1.4 mV.

Exercise 7.6: Twin-T notch

a) Dimension a twin-T notch for 50 Hz with $C = 100$ nF.

b) Its transfer function is $\underline{H} = \dfrac{1 - x^2}{1 - x^2 + j4x}$ with $x = f/f_0$. Calculate $|H|$ at 5 Hz and at 50.5 Hz.

c) Compare with the first-order low-pass with $f_c = 2.34$ Hz. Which filter would you use in the capstone?

d) Check a) and b) in Falstad.

Solution 7.6

a) $R = 1/(2\pi \cdot 50\,\text{Hz} \cdot 100\,\text{nF}) = 31.8\,\text{k}\Omega$, $R/2 = 15.9\,\text{k}\Omega$, $2C = 200\,\text{nF}$.

b) 5 Hz: $x = 0.1$, $|H| = 0.99/\sqrt{0.99^2 + 0.4^2} = 0.927$.
50.5 Hz: $x = 1.01$, $|H| = 0.0201/\sqrt{0.0201^2 + 4.04^2} = 0.0050$.

c) At 50 Hz the notch is much better (-46 dB at 50.5 Hz and more at exactly 50 Hz), but it attenuates the signal at 5 Hz as well, it needs precise components, and it does nothing against other frequencies above $f_s/2$. The anti-aliasing low-pass is required anyway; the notch is at most an addition.

d) ▶ Falstad: `examples/filters/filter_twin_t.txt`
`Vout_5Hz` $= 0.927$ V, `Vout_50Hz` $= 0.45$ mV (input 1 V).

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